Word in noise - Insight46 phase 1

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Document details

Categories of variables	Imaging outcome measure – normal appering white mtter (NAWM) diffusion tensor imaging (DTI) parameters indexing microstructural integrity including: 1. fractional anisotropy (FA) 2. mean diffusivity (MD) 3. neurite orientation dispersion and density imaging (NODDI) derived parameters of neurite density index (NDI) 4. neurite orientation dispersion and density imaging (NODDI) derived parameters of orientation dispersion index (ODI) Alterations of NAWM integrity can be assessed by diffusion tensor imaging (DTI) which enables sensitive estimation of white matter tissue microstructure environments using water diffusion properties. Parameters typically include fractional anisotropy (FA), a measure of directionality, and mean diffusivity (MD), a measure of magnitude of diffusion. Additional multicompartmental modelling techniques have been developed to address limitations in these conventional measures. For example, neurite orientation dispersion and density imaging (NODDI) estimates microstructural complexity parameters of neurite density index (NDI), a measure of fraction of axons and dendrites; and orientation dispersion index (ODI), a measure of variability of neurite orientation (5). Lower values for FA and NDI and higher values for MD and ODI are thought to represent worse microstructural integrity.
Summary of work undertaken	Cleaning± **Description/Rationale:** DTI cleaning pipeline requires: QC'd DTI scans: shell b700 (for FA, MD) and shell b2000 (for NODDI ODI FICVF) To generate NAWM metrics requires: QC'd DTI scans QC'd T1, T2 and FLAIR BaMoS final WMH masks FA, MD and NODDI masks **Values:** Mean NAWM values for FA, MD, ODI and FICVF were generated.
Names of source variables	is, the, only, variable, of, interest, that, is, extracted, from, these, source, files, for, the, final, datafile
Output variables	qc_nawm_all_i46p1, qc_nawm_all_noddi_i46p1, qc_nawmreason_i46p1, nawmz_fa_f_i46p1, nawmz_md_f_i46p1, nawmz_ficvf_f_i46p1, nawmroa_fa_f_i46p1, nawmroa_md_f_i46p1, nawmz_odi_f_i46p1, nawmroa_ficvf_f_i46p1, nawmroa_odi_f_i46p1
Papers using these variables	In progress

Background

Alterations of NAWM integrity can be assessed by diffusion tensor imaging (DTI) which enables sensitive estimation of white matter tissue microstructure environments using water diffusion properties. Parameters typically include fractional anisotropy (FA), a measure of directionality, and mean diffusivity (MD), a measure of magnitude of diffusion. Additional multicompartmental modelling techniques have been developed to address limitations in these conventional measures. For example, neurite orientation dispersion and density imaging (NODDI) estimates microstructural complexity parameters of neurite density index (NDI), a measure of fraction of axons and dendrites; and orientation dispersion index (ODI), a measure of variability of neurite orientation (5). Lower values for FA and NDI and higher values for MD and ODI are thought to represent worse microstructural integrity.

DTI NAWM: reason failed	Freq.	Percent	Cum.
pass	411	81.87	81.87
T1 fail	3	0.6	82.47
FLAIR fail	3	0.6	83.07
BAMOS segmentation fail	1	0.2	83.27
cortical stroke mis-segmented	5	1	84.26
demyelination mis-segmentated	3	0.6	84.86
vascular anommaly mis-segmented	1	0.2	85.06
no scan	31	6.18	91.24
short scan protocol (no dti)	17	3.39	94.62
QC DTI B700 fail	14	2.79	97.41
QC DTI B2000/noddi fail	13	2.59	100

	Freq.	Percent	Cum.
in46_w1_qc_nawm_all			
0 = didn't pass	78	15.54	15.54
1 = pass	424	84.46	100

	Freq.	Percent	Cum.
in46_w1_qc_nawm_all_noddi			
0 = didn't pass	91	18.13	18.13
1 = pass	411	81.87	100

Variables in this document

11 high-confidence matches

Variable	Label	How it was found
Sub-Studies [57] › Insight46 [699] › Brain MRI [6033] › Normal appearing white matter (NAWM) [60333]		
nawmroa_fa_f_i46p1	QC'd normal appearing white matter (nawm) mean fractional anisotropy (fa) - raw value - Cross-sectional - Insight46 phase 1	topsheet field
nawmroa_ficvf_f_i46p1	QC'd normal appearing white matter (nawm) mean intracellular volume fraction (ficvf) raw value - Cross-sectional - Insight46 phase 1	topsheet field
nawmroa_md_f_i46p1	QC'd normal appearing white matter (nawm) mean diffusivity (md) - raw value - Cross-sectional - Insight46 phase 1	topsheet field

Variable	Label	How it was found
nawmroa_odi_f_i46p1	QC'd normal appearing white matter (nawm) mean orientation dispersion index (odi) - raw value - Cross-sectional - Insight46 phase 1	topsheet field
nawmz_fa_f_i46p1	QC'd normal appearing white matter (nawm) mean fractional anisotropy (fa) - standardised to control sample - Cross-sectional - Insight46 phase 1	topsheet field
nawmz_ficvf_f_i46p1	QC'd normal appearing white matter (nawm) mean intracellular volume fraction (ficvf) - standardised to control sample - Cross-sectional - Insight46 phase 1	topsheet field
nawmz_md_f_i46p1	QC'd normal appearing white matter (nawm) mean diffusivity (md) - standardised to control sample - Cross-sectional - Insight46 phase 1	topsheet field
nawmz_odi_f_i46p1	QC'd normal appearing white matter (nawm) mean orientation dispersion index (ODI) - standardised to control sample - Cross-sectional - Insight46 phase 1	topsheet field
qc_nawm_all_i46p1	Diffusion tensor imaging (DTI) all reasons: 0 (don't use) 1 (use) - Cross-sectional - Insight46 phase 1	topsheet field
qc_nawm_all_noddi_i46p1	Diffusion tensor imaging (DTI) neurite orientation dispersion and density imaging (NODDI) all reasons: 0 (don't use) 1 (use) - Cross-sectional - Insight46 phase 1	topsheet field
qc_nawmreason_i46p1	Diffusion tensor imaging (DTI) normal appearing white matter (NAWM): reason failed - Cross-sectional - Insight46 phase 1	topsheet field

1 medium-confidence match

Variable	Label	How it was found
Questionnaire data [55] › Health and medical history [510] › Medical conditions [5101] › Heart and circulatory conditions [51014]		
stroke	Have you had a stroke in the last 10 years - at age 53 years	prose scan